1. Starting points

Axioms

1. **Axiom.** Attending a WG21 meeting in person costs money and time.
2. **Axiom.** A sponsor can pay a participant's travel and salary, but cannot attend, speak, or vote in the participant's place.
3. **Axiom.** A participant who attends most meetings expects to meet many of the same colleagues at future meetings. A chair expects fewer problems from advancing a paper whose author has a record of following the procedures and responding to review.
4. **Axiom.** Given a choice, and other things equal, each participant, sponsor, national body, and user prefers the option that brings it more of what it values, sooner, at less cost to it, and weighs a gain in one value against a loss in another on one scale of its own. For this proof, a participant values adoption of the proposals they support, the interests of their sponsor if they have one, their standing, keeping any chair or convenor post they hold, their time and money, and a good result for users; a sponsor values its interests, which can depend on particular features of a design; a national body values the outcomes its national position supports and, where it has none yet, the interests concerned at its national level; and a user values the features they use or would use, how well those features work, and their own time and money.
5. **Axiom.** A participant who votes Against a colleague's proposal cannot rule out that the colleague will support the participant's own proposals less readily. A Neutral vote, or not voting, gives the colleague no more reason for this than an Against vote does.
6. **Axiom.** No participant can study every paper closely, so each chooses which papers to study, and what a participant learns from a paper or its discussion can change their vote.
7. **Axiom.** Writing or reviewing a paper takes time, and, other things equal, a longer or more complex paper takes more; preparing to give reasons for a vote also takes time.
8. **Axiom.** Until a design is implemented and used, nobody can be sure how well it works, and even then those voting on it cannot fully judge how well it serves users. A design with more parts and more interactions with existing wording has more places where a defect can go unnoticed.
9. **Axiom.** When a proposal is first presented, it has received less committee review than at any later meeting. When a design is presented in parts, reviewers of an early part cannot study in detail what the later parts will require of it until those parts are presented.
10. **Axiom.** If a decision is reversed, the work built on it must be redone or discarded. After a favorable poll, authors and others build more on the design, such as wording, implementations, and proposals that depend on it.
11. **Axiom.** Revising a design throughout costs its authors more work than a change confined to one part of it.
12. **Axiom.** A participant learns how the committee works partly by attending, some of it only that way, and this knowledge differs from skill at design. Participants can use only procedural options they know,

and, other things equal, those who know more spend less effort getting a paper considered and have a better chance of adoption.

13. **Axiom.** Over the years, some participants stop attending and others join, while a published paper stays available to later readers for decades. Someone absent from a meeting learns what was said there only from writing or from those present who remember it, and a clear written objection is easier to raise again.
14. **Axiom.** A meeting has a fixed number of hours, so time given to one paper is not available to another.
15. **Axiom.** WG21 plans a new edition of the C++ standard every three years, and ISO publishes each edition once national bodies approve it.
16. **Axiom.** When several subgroups meet at the same time, a participant can attend only one of them.
17. **Axiom.** A design in the standard applies to every user of its feature, so a better design's gain is spread across them all, while participating costs the participant or their sponsor. Users differ from each other and from participants in the features they care about and what they can spend taking part, and one user's effort seldom changes a committee outcome.

Definitions

18. **Definition.** A participant is a registered expert attending a WG21 meeting. Guests a convenor invites are not participants.
19. **Definition.** A regular is a participant who expects to attend future meetings with many of the same colleagues.
20. **Definition.** A newcomer is a participant attending for the first time; a newcomer who expects to attend future meetings with many of the same colleagues is also a regular.
21. **Definition.** An author is a participant who presents a proposal paper as one of its authors.
22. **Definition.** A sponsor is an organization that pays some or all of a participant's costs of taking part.
23. **Definition.** A user is a person who writes C++ and does not participate.
24. **Definition.** A design room is EWG or LEWG, a study group is any WG21 study group, a wording room is CWG or LWG, and a subgroup is any of these. Plenary is a WG21 meeting's closing plenary session, a chair chairs a subgroup, the convenor convenes WG21, and a consensus call is a chair's or the convenor's determination of consensus.
25. **Definition.** A national body is an ISO member body that is a P-member of SC22, the committee WG21 belongs to. In quotations from the Directives, "the committee" is SC22, "the Chair" its chair, and "National Body" any ISO member body, which on enquiry and final ballots includes non-P-members of SC22.
26. **Definition.** Favor, Neutral, and Against are the poll positions as SD-4 capitalizes them, with Strongly Favor and Weakly Favor counting as Favor and likewise for Against. The 2:1 norm is SD-4's statement on subgroup polls that a proposal normally advances when, after discussion of the concerns of those voting against, more than twice as many are in favor as against, even if many vote Neutral, though a Neutral majority of those voting can be concerning.

27. **Definition.** A participant's standing is the readiness of colleagues to support the participant's future proposals.
28. **Definition.** A player's stake in a poll is the difference the poll's outcome makes to what the player values.
29. **Definition.** A statement that a player does better with one option than another means the option is worth more to the player, by their weighing of the values the statement names. Every value the statement does not name is held equal.
30. **Definition.** A direction poll is a poll on whether to continue work on a design, and a forwarding poll is a poll to send a proposal to the next room.
31. **Definition.** An incumbent on a topic is a design that has passed a direction poll in a design room and has not failed a later direction poll there.
32. **Definition.** An edition is one published version of the C++ standard, and its feature cutoff is the last meeting at which a new feature can be approved for it.
33. **Definition.** A reversal is the removal or replacement of a design or decision that WG21 or one of its subgroups has approved.
34. **Definition.** A player is a person or body whose choices the proof considers. A settled pattern, which game theory calls a Nash equilibrium, is a set of choices, one per player, in which no player gains by changing only their own choice.
35. **Definition.** A small proposal is a proposal by one author who pays their own costs and expects its design to be reviewed within one meeting.
36. **Definition.** A large proposal is a proposal whose two or more authors have their costs paid by one or more sponsors and expect its review to span more than one edition. Its paper is longer and more complex than the paper of any small proposal.
37. **Definition.** A result-seeking author is an author whose gain from a proposal depends on how well the adopted design serves its users, not on whether it keeps any particular feature.
38. **Definition.** A design-bound author is one whose gain from a proposal depends on the adopted design keeping particular features fixed before review, for example because a sponsor's interests depend on them.

Facts

39. **Fact.** SD-4's section "Purpose and audience" contains these statements.
    a. **Fact.** "This document summarizes some of WG21's current practices and procedures, in addition to the requirements of the ISO/IEC Directives and JTC 1 Supplement available at the ISO Directives and Policies page." "Note: This document does not repeat the basic material already covered in the following:" (SD-4, "Purpose and audience")

b. **Fact.** "Although this document is also useful to new participants to get up to speed, the material herein covers important procedures and situations that arise regularly in our meetings, email lists, formal ballots, and other participation, and everyone who participates in WG21 is expected to be familiar with this information." (SD-4, "Purpose and audience")

40. **Fact.** SD-4's section "Code of Conduct" contains these statements, all but the first from a slide it displays.
    a. **Fact.** "To report a concern about a potential Code of Conduct violation, follow the steps in the ISO document “Guidance and Process For Addressing Misconduct and Breaches of the Code of Conduct.”" (SD-4, "Code of Conduct")
    b. **Fact.** "Act in good faith & with due care & diligence" (SD-4, "Code of Conduct")
    c. **Fact.** "Avoid collusive or anticompetitive behavior" (SD-4, "Code of Conduct")
    d. **Fact.** "Respect others & their opinions" (SD-4, "Code of Conduct")
    e. **Fact.** "Accept group decisions" (SD-4, "Code of Conduct")
    f. **Fact.** "Ensure that all views are heard & understood" (SD-4, "Code of Conduct")
41. **Fact.** SD-4's section "Consensus" contains these statements.
    a. **Fact.** "consensus: General agreement, characterized by the absence of sustained opposition to substantial issues by any important part of the concerned interests and by a process that involves seeking to take into account the views of all parties concerned and to reconcile any conflicting arguments." "NOTE Consensus need not imply unanimity." (SD-4, "Consensus")
    b. **Fact.** "Consensus is not a tyranny of the majority. The majority cannot just say “we have enough votes” and refuse to listen to the concerns of the minority. The ISO Code of Conduct requires the majority to 'ensure that the views of all are heard and understood.'" (SD-4, "Consensus")
    c. **Fact.** "Consensus is not a tyranny of the minority. The minority, once they have had opportunity to be heard, cannot just say “we still object so you can't proceed” and refuse to accept the consensus of the majority to pursue a different direction. The ISO Code of Conduct requires the minority to 'accept group decisions.'" (SD-4, "Consensus")
    d. **Fact.** "Consensus does not preclude judicious escalation (“throwing an exception”). In rare cases, where the minority still believe that a serious error of judgment is being made, escalation can be appropriate in accord with the Escalation procedures covered later in this document; note that the key words are “rare” (not regular), “still” (it was raised before), and “serious.”" (SD-4, "Consensus")
    e. **Fact.** "Escalation can become inappropriate in two ways: (a) when a serious objection is not raised “in a timely manner,” such as being withheld until plenary session or formal ballot time rather than being raised consistently throughout the process starting as early as possible; or (b) when a participant or national body regularly uses the escalation process to express a pattern of strong disagreement on topic after topic, which erodes their credibility and is not the purpose of the escalation resolution process (like exception handling, escalation handling is for hard errors, and is not designed for expressing less serious conditions or for what should be ordinary control flow)." (SD-4, "Consensus")
42. **Fact.** SD-4's section "Proposals and papers" contains these statements.

a. **Fact.** "If a proposal doesn't have a paper, it doesn't exist. Any significant proposal requires a paper, including every proposal that is a design addition or change." "Note: A paper is not needed for minor issues that are about improving standardese wording to implement the decided design. Those do not affect the design itself, and can be put on issues lists only." (SD-4, "Proposals and papers")
b. **Fact.** SD-4's list under "A good proposal paper should:" includes "Be principle-based: Articulate the design principles for the proposed solution. Show how the proposed solution fits with the rest of the language's principles and design philosophy (note: explicitly not with the language's quirks)." (SD-4, "Proposals and papers")
c. **Fact.** SD-4's list under "A good proposal paper should:" includes "Show design alternatives considered: along with concrete examples showing why they were not pursued." (SD-4, "Proposals and papers")
d. **Fact.** "All papers to be considered at a meeting must be received before that meeting's pre-meeting mailing deadline. The agenda for each meeting includes all papers in mailings since the previous meeting (from the post-meeting mailing of the previous meeting to the pre-meeting mailing of this meeting, inclusive), as well as possibly selected earlier papers that were not sufficiently addressed at a previous meeting in the judgment of the subgroup chair." (SD-4, "Proposals and papers")
e. **Fact.** "Corollary: If a topic does not have at least one on-time paper, the whole topic is not on the meeting agenda." "Per ISO rules, if a WG meeting were to take action on a topic that was not mentioned on the agenda (including in the topics of on-time papers), a national body could formally escalate an objection in SC 22 and JTC 1 to the entire WG meeting on the grounds that the national body had insufficient notice so as to be able to send any (or the right) participants to participate in that technical discussion topic." (SD-4, "Proposals and papers")
f. **Fact.** "A proposal that does not have an on-time paper published since the previous meeting is not actionable and will not be considered during official meeting time; the author is encouraged to submit a revision as an on-time paper for a future meeting." (SD-4, "Proposals and papers")
g. **Fact.** "Followup papers to an on-time paper, such as late or in-meeting rebuttal/elaboration/update papers to an on-time paper, can of course be considered; they are part of considering the on-time paper." (SD-4, "Proposals and papers")
h. **Fact.** "A proposal that does not have one of its authors, or another qualified and prepared presenter, to present it is not actionable and will not be considered during official meeting time." (SD-4, "Proposals and papers")
i. **Fact.** "The purpose of a presentation is for a person who is an expert in the paper's domain (including in constraints as well as design alternatives considered) to educate a group of general C++ experts." (SD-4, "Proposals and papers")
j. **Fact.** "The presenter must be someone who is invested in the proposal and will ensure feedback will be used, which means the presenter will personally produce a new revision of the paper based on feedback or else will communicate feedback to the author to do so." (SD-4, "Proposals and papers")

k. **Fact.** "We cannot act on ideas without papers, and we do not significantly delay progress on concrete proposals in order to wait for alternative proposals we might get in the future. If we have an active proposal that is making progress, and an objection is raised that a different competing approach would be better but that other approach does not yet have a paper, the EWG or LEWG subgroup chair may elect to delay the progress of the active proposal by one meeting to give the objector(s) opportunity to bring an on-time proposal paper. Otherwise, if a competing alternative does not have a paper, it does not exist and will not block progress of a proposal that we do have before us." (SD-4, "Proposals and papers")

l. **Fact.** "At early discussion phases, subgroup chairs have discretion to allow limited exploration of alternatives to an active proposal to decide whether a second paper is warranted, or to discuss broader prerequisite topics that are exposed by a paper under discussion." (SD-4, "Proposals and papers")

m. **Fact.** "If a proposal has consensus to be in a TS, but does not yet have consensus to be in the IS, that means its progress is delayed by approximately 1-2 years; that is the time it takes to publish the TS, and then wait for usually one year before the TS is proposed to be merged into the IS with any changes based on the TS experience." (SD-4, "Proposals and papers")

n. **Fact.** "If by the time the TS is ready to be considered for merging into the IS we do not have alternative proposals actively progressing, the default action to is to move forward with the proposal we have, we will not wait indefinitely for something better." (SD-4, "Proposals and papers")

43. **Fact.** SD-4's section "Subgroup responsibility and polls" contains these statements.

a. **Fact.** "Subgroup chairs are appointed by the convener, and are selected to match the current needs of the subgroup. They have no fixed term." (SD-4, "Subgroup responsibility and polls")

b. **Fact.** "In subgroup polls, each person in the room can cast one vote." (SD-4, "Subgroup responsibility and polls")

c. **Fact.** "The subgroup chair may take any polls they choose." (SD-4, "Subgroup responsibility and polls")

d. **Fact.** "Multiple polls, where multiple alternatives are polled (e.g., by 5-way poll) in order to see which has most consensus. This is encouraged so as not to lose the option that has the most consensus because of the order in which polls are taken." (SD-4, "Subgroup responsibility and polls")

e. **Fact.** "Subgroup polls, especially in design subgroups, should favor progress." (SD-4, "Subgroup responsibility and polls")

f. **Fact.** "A proposal normally advances if there are more than twice as many in favor of a proposal as against, after discussion of the concerns of those voting against and possibly a re-poll to see if opinions have improved. This is true even if a large number vote Neutral, though it can be concerning if a majority of all those voting vote Neutral." (SD-4, "Subgroup responsibility and polls")

g. **Fact.** "Not everyone in the room needs to vote (even Neutral). Not everyone is interested in every proposal, and participants who are not familiar with the material of a particular poll (did not read the paper and/or were not closely paying attention to the discussion) typically do not vote on that poll." (SD-4, "Subgroup responsibility and polls")
h. **Fact.** "Anyone voting Against, especially Strongly Against, may be asked for their reasons by the chair. They should be prepared to articulate their rationale, including either what change(s) would address their concern and change their position, or else that they are opposed to pursuing the proposal in any form and no change would change their position." (SD-4, "Subgroup responsibility and polls")
i. **Fact.** "The direction group maintains a list of the proposals it considers the most important for the next version of C++ or to otherwise make progress such as in a TS, and the design group chairs use that that list to prioritize work at meetings." (SD-4, "Subgroup responsibility and polls")
j. **Fact.** Of the direction group, SD-4 says "Their charter includes setting forth a group opinion on:" and lists "Providing an opinion on any specific proposal: This includes whether the proposal should be pursued or not, and if pursued any changes that should be considered. Design group participants are strongly encouraged to give weight to an opinion that the design group feel strongly enough about to suggest." (SD-4, "Subgroup responsibility and polls")
k. **Fact.** "Study groups (SGs) own “incubating” to produce an initial proposal that can then enter the usual design group processing." (SD-4, "Subgroup responsibility and polls")
l. **Fact.** "An SG is formed by the convener at the recommendation of a design subgroup chair when there appears to be broad interest in a topic, there is not yet a unified initial proposal to bring to the design subgroup (typically because proposals are immature, incomplete, and/or competing), and there is a strong candidate to chair the SG to ensure it operates productively." (SD-4, "Subgroup responsibility and polls")
m. **Fact.** "The design subgroups (EWG and LEWG) own selecting and recommending feature designs and ship vehicles (TS or IS)." (SD-4, "Subgroup responsibility and polls")
n. **Fact.** "Both positive and negative comments are welcome, but should always be constructive and respectful." (SD-4, "Subgroup responsibility and polls")
o. **Fact.** "If anyone has a concern that a particular new proposal should not take up extended subgroup time (e.g., is not well thought out, is in a direction that previously did not get interest or consensus, or is otherwise unlikely to progress in its current form), they should share their concern privately and in advance with the subgroup chair." (SD-4, "Subgroup responsibility and polls")
p. **Fact.** "The subgroup chair may choose (based on such requests or just on their own judgment) to expedite the proposal's initial handling to make sure a representative overview is heard but then determine whether it has enough subgroup interest before spending extended time exploring the details; for example, the chair might time-box initial discussion and then poll whether the subgroup is interested in discussing the proposal's details further." (SD-4, "Subgroup responsibility and polls")

q. **Fact.** "The proposal author is expected to participate in the design subgroup to present the proposal, answer questions, participate in design discussion, and implement any changes directed by the group, typically for subsequent revisions of the proposal to be considered at the next meeting. In rare cases, if the proposal author chooses not to implement a change directed by the group, the author should be prepared to defend that decision and bring new information to the next meeting as part of the paper revision; proposals that do not implement design subgroup direction harm their own chances of gaining consensus within the design group and/or in plenary." (SD-4, "Subgroup responsibility and polls")

r. **Fact.** "When a proposal's design achieves a general consensus of the design subgroup, as determined by the subgroup chair, for its design and ship vehicle, it is forwarded to the specification subgroups." (SD-4, "Subgroup responsibility and polls")

s. **Fact.** "The specification subgroups (CWG and LWG) own the recommendation of the “standardese” specification (aka “wording”) that implements the intended design." (SD-4, "Subgroup responsibility and polls")

t. **Fact.** "If the specification subgroup has design questions or suggestions, those questions are referred back to the design group." (SD-4, "Subgroup responsibility and polls")

u. **Fact.** "When a proposal's specification achieves a general consensus of the specification subgroup, as determined by the subgroup chair, that its specification correctly expresses the intended design, it is forwarded to the closing plenary (whole-WG21) session." (SD-4, "Subgroup responsibility and polls")

v. **Fact.** "Subgroup chairs prioritize first papers that are related to one or more of the following: Direction priorities recommended in" ... "work the committee is aiming to include in the next IS; and work that can make plenary progress at this meeting. After that they get to things on a best-effort basis as time allows." (SD-4, "Subgroup responsibility and polls"; the omitted words are a paper number)

44. **Fact.** SD-4's section "Closing plenary session responsibility and polls" contains these statements.

a. **Fact.** "Design subgroup chairs present the proposals whose designs were approved at this meeting, including to solicit all present to express any serious design concerns that may not have been discovered in subgroups so that these can be known and addressed before extensive wording effort is invested." (SD-4, "Closing plenary session responsibility and polls")

b. **Fact.** "The default is to ask if there is objection to unanimous consent. (Unanimous consent means that all participant positions are either Favor or Neutral, and that none are Against). Most polls pass by unanimous consent." (SD-4, "Closing plenary session responsibility and polls")

c. **Fact.** "If there is an objection to unanimous consent, the Convener takes a 3-way poll as described above. If there is significant opposition, the Convener will usually ask whether the Against votes are personal positions or national body positions, typically by asking the national body panel chairs who are present." (SD-4, "Closing plenary session responsibility and polls")

d. **Fact.** "The WG21 closing plenary session owns approval of adding a proposal and its specification wording to a working draft (TS or IS). The plenary session's responsibility is to decide whether

there is consensus, as determined by the Convener, to adopt each proposed change that is recommended by the design and specification subgroups." (SD-4, "Closing plenary session responsibility and polls")

e. **Fact.** "The plenary session does not actively change the design or entertain extended design discussion, which always happens in the design subgroups." (SD-4, "Closing plenary session responsibility and polls")

f. **Fact.** "However, the plenary session participants may ask questions like: “does this handle case X?” and “did you consider alternative Y?” For such questions, the meeting chair will ask the appropriate subgroup chair (who may invite comment from the proposal's author) to answer whether that item was considered and to summarize the result of the discussion and the reason for the decision of the subgroup." (SD-4, "Closing plenary session responsibility and polls")

45. **Fact.** SD-4's section "Technical Specifications (TS) and White Papers" contains these statements.

a. **Fact.** "A TS or white paper is an experimental branch. In ISO, a TS or white paper is for a feature that does not yet have consensus for an IS. A TS or white paper is a separately published “experimental branch” that can ship separately to gain experience — either with an experimental part of the design, or to get enough overall experience so that participants are comfortable that the overall design works as well as hoped. We usually do not put work into a TS or white paper unless WG21 likes the general direction the feature is exploring, and would entertain merging it into the IS in some form (possibly after significant further transformation) if it turns out to work out successfully; features we know we do not intend to put in the IS are not worth shipping in experimental form either." (SD-4, "Technical Specifications (TS) and White Papers")

b. **Fact.** Among the ways SD-4 says a TS or white paper differs from a feature in a draft standard, it "asks implementations/users to give feedback to identify shortcomings and puts them on notice to expect breaking changes (they typically should not rely on the feature in production code)." (SD-4, "Technical Specifications (TS) and White Papers")

c. **Fact.** SD-4 describes "pre-merge issues, where we do not have a consensus to merge into the IS working draft until this specific set of issues is first addressed." and says "Issues that people and national bodies were willing to hold in abeyance with provisional resolutions for the TS or white paper are expected to have to be addressed, possibly pre-merge; however, people are encouraged not to rehash decisions that already had consensus when the TS or white paper was started unless new information has emerged." (SD-4, "Technical Specifications (TS) and White Papers")

d. **Fact.** Of the pre-merge issues, SD-4 says "We encourage all participants to keep the second set as small as possible, for real make-or-break concerns only, so that we can merge a TS or white paper early to permit better “integration testing” as we refine it further in the full context of the IS, and permit other parts of the standard to adapt to the new feature." (SD-4, "Technical Specifications (TS) and White Papers")

e. **Fact.** "TS vs white paper: They are functionally the same, but a white paper has a very-lightweight ISO process and is now recommended by ISO and SC22." (SD-4, "Technical Specifications (TS) and White Papers")

f. **Fact.** "A white paper is adopted within our immediate parent committee SC22; it has no prescribed time schedule, no prescribed format, no ISO editing, and when technically complete is sent by WG21 to SC22 to be approved by a simple vote by SC22 national bodies to adopt the draft as a white paper." (SD-4, "Technical Specifications (TS) and White Papers")

g. **Fact.** "Once a feature is in an IS, it is essentially cast in stone and much more difficult to update with breaking changes." (SD-4, "Technical Specifications (TS) and White Papers")

h. **Fact.** "The TS or white paper should list known specific questions on which feedback is desired, including:" "What are the exit criteria (especially, the list of must-address controversial issues) before we can consider merging this TS or white paper into the IS?" (SD-4, "Technical Specifications (TS) and White Papers")

46. **Fact.** SD-4's section "Working draft handling" contains these statements.

a. **Fact.** "In the opening plenary session of each meeting, we routinely adopt the updated working drafts generated after the previous meeting as described in the next paragraph. During the meeting, the wording for all new proposed changes is written against these adopted drafts as the meeting's baseline." (SD-4, "Working draft handling")

b. **Fact.** "After the meeting, each project editor produces an updated candidate working draft that applies all the changes approved for that draft, and generates an editor's report summarizing those changes and any other comments that arose while applying and merging them." "If a working draft is also ready to be sent out for ISO ballot, and there were changes approved to the draft, then the poll to direct the convener to send the document out for ballot also includes naming an editing committee to verify that all approved changes were applied correctly by the project editor before the document is transmitted for ballot." (SD-4, "Working draft handling")

47. **Fact.** SD-4's section "Ballots and comments" contains these statements.

a. **Fact.** "National bodies are expected to send experts to participate in person in WG21 if they want to affect the direction of the standard." (SD-4, "Ballots and comments")

b. **Fact.** "A Disapproval (No) vote should be used only to express that the national body does not want the work (TS or IS) to progress at all in any form — to block further progress on all parts of the work item." (SD-4, "Ballots and comments")

c. **Fact.** "Therefore, No should mean “No, we don't want this TS or IS at all,” because sufficient No votes have that effect — most ballots fail if more than 1/3 of NBs vote No, so voting No only to try to give the NB's comments a higher priority is ineffective (we consider all NB feedback at all times, not just on ballots, and regardless of the Yes/No vote on a ballot) and counterproductive (could actually cause the whole project to fail)." (SD-4, "Ballots and comments")

d. **Fact.** "Note: Ballot comments are treated by ISO, and therefore by WG21, as national body positions." (SD-4, "Ballots and comments")

e. **Fact.** "All editorial and technical comments on the content of that document are in scope. A comment is expected to contain a proposed resolution with sufficient detail to be actionable; if more detail is required than fits easily in the comment form, referencing or attaching a paper is encouraged." (SD-4, "Ballots and comments")

f. **Fact.** "A ballot comment that requests adding an additional feature that is not already in the document is out of scope. New feature proposals are encouraged, but the correct way to bring them to WG21 is by presenting a proposal paper and working to build consensus to add the feature." (SD-4, "Ballots and comments")

g. **Fact.** "A ballot comment that requests a change that was already considered and decided otherwise at a WG21 meeting, and comes from a national body that was present at the meeting and had an opportunity to have their objections be heard and considered, is out of harmony with the ISO Code of Conduct's commitment to 'accept group decisions.' Once the WG has consensus to send a document for ballot, to repeat as an NB comment an objection that previously failed to carry the day is actually making, not a new technical objection, but an objection to the consensus of the WG." "Note: There are two exceptions:" (SD-4, "Ballots and comments")

h. **Fact.** The first exception: "If sufficient new information came to light since the last discussion that it is reasonable to believe enough people could change their opinions to affect the previous consensus. That information, of course, requires an on-time paper for the ballot resolution meeting, and ideally that paper would be available early and referenced directly in the ballot comment." (SD-4, "Ballots and comments")

i. **Fact.** The second exception: "If the national body really strongly objects to a specific item so much that they are willing to vote No on progressing the whole TS or IS that it's in, and have informed the committee all along of the strength of their feeling (see “Escalation…” below). If WG21 were to ballot a document over those objections, they would know that a No With Comments could happen; but the No vote, and negative comments, would be regarding a previously discussed and decided matter, they should never arise for the first time as a surprise No vote on a ballot." (SD-4, "Ballots and comments")

j. **Fact.** "In the case of a successful final approval ballot, where after ballot comment resolution the document will be sent directly to ISO for publication without another ballot, any design change made between the ballot and publication will be expected to have near-unanimous consent in subgroups and in plenary." (SD-4, "Ballots and comments")

48. **Fact.** SD-4's section "How to report a potential technical concern, at any level of severity" contains these statements.

    a. **Fact.** "Describe the problem, and ask whether it was considered. If the concern is about a feature proposal (whether already adopted or still underway) include the proposal authors; as the feature's designers they will be among the best experts about the issue." (SD-4, "How to report a potential technical concern, at any level of severity")

    b. **Fact.** "As early as possible, inform the EWG and LEWG chairs and the proposal's authors: They need to be aware of your concern, are trained in triaging and prioritizing issues and concerns, and can offer calm advice on how to proceed." "As early as possible, inform the subgroup: preferably during the discussion in a meeting, even more preferably before a meeting on a reflector." (SD-4, "How to report a potential technical concern, at any level of severity")

49. **Fact.** SD-4's section "Escalating a serious “cannot live with” objection, or significant new information" contains these statements.
    a. **Fact.** "As early as possible, make the EWG and LEWG chairs aware of your concern. If it is a subgroup decision made during the week and that will be brought forward to plenary that week, you should proactively raise it during the week:" "On the committee email lists: Ideally the issue should be raised on the relevant subgroup's email list immediately following the subgroup discussion to which the objection appertains, but in no case any later than the deadline of 5pm the evening before the closing plenary session (and posting much sooner than that is strongly encouraged, ideally right after the subgroup decision they feel is wrong, is strongly encouraged)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    b. **Fact.** "Within your own national body: This is so that the national body chair has opportunity to caucus with their members and determine whether the position may be a national position (normally the default is no, not a national position, unless it was discussed before or during the meeting to determine that it is a national position)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    c. **Fact.** "Objections that are not so escalated, but are raised or re-raised in plenary, should not be given weight in plenary." "(In the rare case of new information that is found 'just the night before' plenary, it has still passed the cutoff for plenary participants to be able to give it informed and prepared consideration in time for plenary, and it is a race condition that we linearize by saying it has failed to be raised before the cutoff, and should be handled per the next bullet including with a followup paper.)" (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    d. **Fact.** "If it is new information found after the deadline (including between meetings), they should raise it:" "On the committee email lists." "As a national body ballot comment. If a participant does not raise the concern within their national body, and succeed in convincing enough people in their caucus to make it a national body position (by whatever internal rules that particular national body panel uses to determine such), it won't become a national position." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    e. **Fact.** "In both cases, it should be raised with a discussion paper containing detailed information (at least a draft paper if a final one is not yet available)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
50. **Fact.** SD-4's section "Other topics" contains these statements.
    a. **Fact.** "The following materials are not publicly available and are always password-protected:" "Documents on which ISO asserts copyright, notably the final text of a TS/IS." "Meeting records of subgroup discussion, meeting wikis, and non-public committee email lists (aka reflectors), which often include personal positions and discussion. It is not allowed to quote from these publicly (e.g., in papers and blog posts) except that the following are allowed: (a) quoting straw poll questions and numeric results; and (b) quoting words or positions attributed to a specific person with that person's prior consent." (SD-4, "Other topics")

b. **Fact.** "All meetings must be minuted. This includes telecons as well as in-person meetings." (SD-4, "Other topics")

51. **Fact.** ISO/IEC Directives Part 1, clauses 0.1 and 0.7, contain these statements.
    a. **Fact.** "This document replaces the ISO/IEC Directives, Part 1:2024, including the ISO, IEC and JTC 1 Supplements." "For the first time in the history of ISO and IEC, a single common set of procedures for the technical work is applicable across ISO and IEC." (ISO/IEC Directives Part 1, 0.1)
    b. **Fact.** "These procedures take into account the total cost of the operation. The concept of “total cost” includes direct expenditure by National Bodies, expenditure by the offices in Geneva (funded mainly by the dues of National Bodies), travel costs and the value of the time spent by Experts in working groups and committees at both national and international level." (ISO/IEC Directives Part 1, 0.7)
    c. **Fact.** "Opportunities for remote participation at face-to-face meetings should be provided to the extent possible (see Clause 5)." (ISO/IEC Directives Part 1, 0.7)
    d. **Fact.** "Consensus, which requires the resolution of substantial objections, is an essential procedural principle and a necessary condition for the preparation of International Standards that will be accepted and widely used. Although it is necessary for the technical work to progress speedily, sufficient time is required before the approval stage for the discussion, negotiation and resolution of significant technical disagreements." (ISO/IEC Directives Part 1, 0.7)
    e. **Fact.** "Similarly, to avoid re-discussion, National Bodies have the responsibility to ensure that their technical standpoint is established, taking account of all interests concerned at the national level, and that this standpoint is made clear at an early stage of the work rather than, for example, at the final (approval) stage." (ISO/IEC Directives Part 1, 0.7)
    f. **Fact.** "National Bodies are encouraged to abstain if there is a lack of national consensus or Expert input on a specific ballot." (ISO/IEC Directives Part 1, 0.7)

52. **Fact.** ISO/IEC Directives Part 1, clauses 1.7, 1.8.2, 1.12.1, and 1.12.2, contain these statements about committees and working groups.
    a. **Fact.** "Nevertheless, all National Bodies, irrespective of their status within a committee, have the right to vote on enquiry drafts (see 2.6) and on final draft International Standards (see 2.7)." (ISO/IEC Directives Part 1, 1.7)
    b. **Fact.** Among the duties ISO/IEC Directives Part 1 gives the Chair of a committee (for WG21, the chair of SC22): "d) ensure at meetings that all points of view expressed are adequately summarized so that they are understood by all present;" and "e) ensure at meetings that all decisions are clearly formulated and made available in written form by the Secretary/Committee Manager for confirmation during the meeting;" (ISO/IEC Directives Part 1, 1.8.2)
    c. **Fact.** "A WG operates by consensus, reports and gives recommendations, if any, to its parent committee through a WG Convenor appointed by the parent committee. The Convenor shall act in a purely international capacity, divesting themselves of a national position." (ISO/IEC Directives Part 1, 1.12.1)

d. **Fact.** "WG Convenors shall be appointed by the committee for up to three-year terms. Such appointments shall be confirmed by their National Body (or liaison). The Convenor may be reappointed for additional terms of up to three years. There is no limit to the number of terms. Prior to the expiry of the term, the Secretary/Committee Manager shall issue a six-week minimum call to identify any candidates who wish to be considered for the position. If multiple candidates are nominated, the committee shall appoint a Convenor from among the candidates." (ISO/IEC Directives Part 1, 1.12.1)
e. **Fact.** "A WG comprises a number of Experts individually appointed by the P-members, category A liaison and Liaison Representatives from committees in liaison (see 1.15 and 1.16) and category C organizations (see 1.17)." (ISO/IEC Directives Part 1, 1.12.1)
f. **Fact.** "The Experts appointed by a P-member or category A liaison act in a personal capacity and not as the official representative of them. The Liaison Representatives appointed by a category C liaison or committees in liaison act as official representatives of the organizations by which they have been appointed." (ISO/IEC Directives Part 1, 1.12.1)
g. **Fact.** "It is recommended that the Experts and Liaison Representatives keep close contact with that P-member or liaison to inform them about the progress of the work and of the various opinions in the WG at the earliest possible stage." (ISO/IEC Directives Part 1, 1.12.1)
h. **Fact.** "It is recommended that WGs be reasonably limited in size. The committee may therefore decide upon the maximum number of Experts appointed by each P-member and liaison organization." (ISO/IEC Directives Part 1, 1.12.1)
i. **Fact.** "Experts not registered to a WG in the ISO GD or the IEC EMS, as appropriate, shall not participate in its work." (ISO/IEC Directives Part 1, 1.12.2)
j. **Fact.** "Convenors may invite a specific guest to participate in up to two meetings to provide relevant expertise to the WG and shall notify the guest's National Body of the invitation ahead of the meeting via the Office of the CEO (copying the Secretary/Committee Manager of the committee)." (ISO/IEC Directives Part 1, 1.12.2)

53. **Fact.** ISO/IEC Directives Part 1, clauses 2.5.6, 2.5.7, 2.5.9, and 2.5.10, contain these statements about consensus and the committee stage.
    a. **Fact.** "Consideration of successive drafts shall continue until a consensus of the P-members as judged by the Chair has been obtained, or a decision to abandon or defer the project has been made." (ISO/IEC Directives Part 1, 2.5.6) "The decision to circulate an enquiry draft (see 2.6.1) shall be taken by the Chair of the committee on the basis of the consensus among the P-members." "In case of doubt concerning consensus on a revised CD, approval by a two-thirds majority of the P-members voting may be deemed to be sufficient for the revised CD to be accepted for registration as an enquiry draft; however, every attempt shall be made to resolve negative votes." (ISO/IEC Directives Part 1, 2.5.7)

b. **Fact.** "It is the responsibility of the Chair of the committee, in consultation with the Secretary/Committee Manager of the committee and, if necessary, the Project Leader, to judge whether there is consensus bearing in mind the definition of consensus given in ISO/IEC Guide 2:2004 [11]." (ISO/IEC Directives Part 1, 2.5.7)
c. **Fact.** "However, “sustained oppositions” are views expressed at minuted meetings of the committee, working group (WG) or other groups (e.g. advisory groups) and which are maintained by an important part of the concerned interest and which are incompatible with the committee consensus." (ISO/IEC Directives Part 1, 2.5.7)
d. **Fact.** "The notion of “concerned interest(s)” will vary depending on the dynamics of the committee and shall therefore be determined by the committee leadership on a case-by-case" ... "basis." (ISO/IEC Directives Part 1, 2.5.7; the omitted text is a page number and copyright line)
e. **Fact.** "The concept of sustained opposition is not applicable in the context of National Body votes on enquiry or approval ballots since these are subject to the applicable voting rules." (ISO/IEC Directives Part 1, 2.5.7)
f. **Fact.** "Those expressing sustained opposition have a right to be heard, and the following approach is recommended when a sustained opposition is declared." "The leadership shall first assess whether the opposition can be considered a “sustained opposition”, i.e. whether it has been sustained by an important part of the concerned interest. If this is not the case, the leadership will register the opposition (i.e. in the report/minutes, records, etc.) and continue to lead the work on the document." (ISO/IEC Directives Part 1, 2.5.7)
g. **Fact.** "If the leadership determines that there is a sustained opposition, it shall try and resolve it in good faith. However, a sustained opposition is not akin to a right to veto. The obligation to address the sustained oppositions does not imply an obligation to resolve them successfully." (ISO/IEC Directives Part 1, 2.5.7)
h. **Fact.** "The responsibility for assessing whether or not a consensus has been reached rests entirely with the leadership. This includes assessing whether there is sustained opposition or whether any sustained opposition can be resolved without compromising the existing level of consensus on the rest of the document. In such cases, the leadership will register the opposition and continue the work." (ISO/IEC Directives Part 1, 2.5.7)
i. **Fact.** "Those parties with sustained oppositions may avail themselves of appeals mechanisms as detailed in Clause 6." (ISO/IEC Directives Part 1, 2.5.7)
j. **Fact.** "The committee stage ends when all technical issues have been resolved, and a CD is accepted for circulation as an enquiry draft and is registered by the Office of the CEO." (ISO/IEC Directives Part 1, 2.5.9) "If the technical issues with an International Standard project cannot all be resolved within the appropriate time limits, committees may consider publishing an intermediate document in the form of a Technical Specification (see 3.1) pending agreement on an International Standard." (ISO/IEC Directives Part 1, 2.5.10)

54. **Fact.** ISO/IEC Directives Part 1, clauses 2.6.3 to 2.6.5, 2.7.2 to 2.7.7, 3.1.2, and 3.1.3, contain these statements about ballots and Technical Specifications.

a. **Fact.** "An enquiry draft is approved if" "a) a two-thirds majority of the votes cast by the P-members are in favour, and" "b) not more than one-quarter of the total number of votes cast are negative." "Abstentions are excluded when the votes are counted, as well as negative votes not accompanied by any comments of a technical nature." (ISO/IEC Directives Part 1, 2.6.3)
b. **Fact.** "If a National Body finds a final draft International Standard unacceptable, it shall vote negatively and state the reasons by submitting comments of a technical nature. It shall not cast a positive vote that is conditional on the acceptance of modifications." (ISO/IEC Directives Part 1, 2.7.2)
c. **Fact.** After an enquiry ballot, one course of action open to the Chair of the committee (for WG21, the chair of SC22) is "a) when the approval criteria of 2.6.3 are met, and no technical changes are to be included, to proceed directly to publication (see 2.8);" (ISO/IEC Directives Part 1, 2.6.4) At the approval stage, "The Secretary/Committee Manager has the responsibility to bring any errors that may have been introduced in the preparation of the draft to the attention of the Office of the CEO by the end of the voting period; further editorial or technical changes are not accepted at this stage." (ISO/IEC Directives Part 1, 2.7.4) "Technical changes to an approved FDIS are not allowed." (ISO/IEC Directives Part 1, 2.7.5)
d. **Fact.** "If the final draft International Standard is not approved in accordance with the conditions in 2.7.3, the document shall be referred back to the committee concerned for reconsideration in light of the technical reasons submitted in support of the negative votes." "The committee shall decide to:" "resubmit a modified draft as a committee draft, enquiry draft or final draft International Standard; or" "publish a Technical Specification (see 3.1); or" "cancel the project." (ISO/IEC Directives Part 1, 2.7.7)
e. **Fact.** "All comments received will be retained for the next review and will be recorded on the voting form as “noted for future consideration”." (ISO/IEC Directives Part 1, 2.7.5)
f. **Fact.** "Not later than 12 weeks after the end of the voting period, a full report shall be prepared by the secretariat of the committee and circulated by the Office of the CEO to the National Bodies. The report shall" "a) show the result of the voting," "b) state the decision of the Chair of the committee," "c) reproduce the text of the comments received, and" "d) include the observations of the secretariat of the committee on each of the comments submitted." (ISO/IEC Directives Part 1, 2.6.5)
g. **Fact.** "Every attempt shall be made to resolve negative votes." "If, within eight weeks from the date of dispatch, two or more P-members disagree with the decision of the Chair, the draft shall be discussed at a meeting (see Clause 5)." "Committees are required to respond to all comments received." (ISO/IEC Directives Part 1, 2.6.5)
h. **Fact.** "A final draft International Standard having been circulated for voting, is approved if" "a) a two-thirds majority of the votes cast by the P-members are positive, and" "b) not more than one-quarter of the total number of votes cast are negative." "Abstentions are excluded when the votes are counted, as well as negative votes not accompanied by any comments of a technical nature." (ISO/IEC Directives Part 1, 2.7.3)

i. **Fact.** "The decision to publish the resulting document as a Technical Specification shall require a two-thirds majority vote of the P-members voting. If technical changes are required following the ballot, a subsequent ballot is required to approve the revised draft." (ISO/IEC Directives Part 1, 3.1.2) "Competing Technical Specifications offering different technical solutions are possible, provided that they do not conflict with existing International Standards." (ISO/IEC Directives Part 1, 3.1.3)

55. **Fact.** ISO/IEC Directives Part 1, clauses 5.3.1, 5.5, and 5.7, contain these statements about meetings.
    a. **Fact.** "Participants shall not comment publicly on the discussions, positions, or contributions made by others during meetings." "Only the meeting leadership may speak publicly on the consensus reached, and only in the context of an official press activity linked to the meeting." (ISO/IEC Directives Part 1, 5.7)
    b. **Fact.** "Meetings hosts shall not charge participants any sort of participation fee, nor require accommodation at specific hotels or hotel rates for any meetings." (ISO/IEC Directives Part 1, 5.3.1)
    c. **Fact.** "The Secretary/Committee Manager or Project Leader/Convenor shall ensure that meeting outcomes are properly documented and shared. Decisions shall be distributed within 48 hours after the meeting, and the meeting report/minutes shall be distributed within four weeks." "If no comments are received within four weeks after distribution, the meeting report/minutes is considered approved. If comments are received, they shall be addressed by the Secretary/Committee Manager or Convenor. If objections are received, the meeting report/minutes remains unconfirmed until the next meeting, and any approved changes are recorded in the meeting report/minutes of that next meeting." (ISO/IEC Directives Part 1, 5.5)
56. **Fact.** ISO/IEC Directives Part 1, clause 6, contains these statements about appeals.
    a. **Fact.** "National Bodies have the right of appeal" "a) to the parent technical committee on a decision of a subcommittee;" "b) to the technical management board on a decision of a technical committee;" "c) to the council board on a decision of the technical management board," "within eight weeks of the decision in question." (ISO/IEC Directives Part 1, 6.1.1)
    b. **Fact.** "A P-member of a committee may appeal against any action, or inaction, on the part of the committee, when the P-member considers that such action or inaction is" "a) not in accordance with" "the Statutes and Rules of Procedure [16], [17];" "the ISO/IEC Directives; or" "b) not in the best interests of international trade and commerce, or such public factors as safety, health or environment." (ISO/IEC Directives Part 1, 6.1.2)
    c. **Fact.** "Matters under appeal may be either technical or administrative in nature." "Appeals on decisions concerning new work item proposals, committee drafts, enquiry drafts and final draft International Standards are only eligible for consideration if" "questions of principle are involved, or" "the contents of a draft may be detrimental to the reputation of ISO or IEC." (ISO/IEC Directives Part 1, 6.1.3)
    d. **Fact.** "When an appeal is against a decision relating to work in progress, the work shall be continued up to and including the approval stage (see 2.7)." (ISO/IEC Directives Part 1, 6.5)

57. **Fact.** INCITS's Organization, Policies and Procedures contain these statements on how INCITS counts votes and forms United States positions.
    a. **Fact.** "There shall be only one voting membership per organization (see Section 3.9.2, Limits on Voting Membership). An alternate representative may vote only if the principal representative fails to vote. Voting by proxy shall not be permitted." (INCITS Organization, Policies and Procedures, 5.2)
    b. **Fact.** The glossary defines "Organization (For the purposes of membership and voting)" as "Corporations, Federal or military departments or agencies, partnerships or associations, self-employed or self-financed individuals, or any other legal or commercial entities." (INCITS Organization, Policies and Procedures, 8.2)
    c. **Fact.** "Note: On TCs, TGs, and SGs, the U.S. Department of Defense (DoD) is allowed a maximum of four votes from separate and unrelated entities such as the Department of Army, Department of Navy, Department of Air Force, Defense Logistics Agency, Defense Mapping Agency, etc. This exception is due to the unique nature of the DoD and is, therefore, not considered a precedent for any further exceptions." "In order to be in the voting member category, consultant organizations shall declare that their participation is not being funded by any organization already having voting membership or by an organization that is not eligible for membership." For standards developing organizations and consortia: "There shall be only one voting membership for each separate organization. For these entities, their principal and alternate representatives may be employed by other organizations that have voting memberships." (INCITS Organization, Policies and Procedures, 3.9.2)
    d. **Fact.** "The appropriate U.S. TAG shall submit a recommendation on the U.S. voting position to the Executive Board. Once the U.S. position is approved by the Executive Board, it shall be submitted to ANSI for transmission to JTC 1 or ISO." (INCITS Organization, Policies and Procedures, 7.11.2) "The INCITS Executive Board is the U.S. TAG to JTC 1." (INCITS Organization, Policies and Procedures, 8.2)
    e. **Fact.** "When a new issue or question comes before a Plenary, or an existing issue or question is modified during the course of the Plenary, the U.S. delegation may determine the official U.S. position by consensus of the U.S. caucus as determined by the HoD." (INCITS Organization, Policies and Procedures, 7.17.1.4)
58. **Fact.** Each item below states an absence that a search of the documents and clauses it names confirms; the edition of the Directives these items search replaces the JTC 1 Supplement, and the proof does not search the other documents SD-4 lists as covering material it does not repeat.
    a. **Fact.** SD-4 contains no requirement that the record of a poll show how an objection was answered or why a consensus call was made (SD-4, whole document; searched: record, minute, reason, answer, tally, objection, why).
    b. **Fact.** SD-4 names no step in which a chair asks voters of any position other than Against for their reasons (SD-4, whole document; searched: reason, rationale, articulate, explain, justif, why).

c. **Fact.** None of ISO/IEC Directives Part 1 clauses 1.8.2, 2.5.7, and 5.5, which govern how decisions, minutes, and oppositions are recorded, requires a record of why consensus was judged to exist (ISO/IEC Directives Part 1, 1.8.2, 2.5.7, 5.5; searched: reason, rationale, justif, why).

d. **Fact.** SD-4 uses the words "sustained" and "concerned interests" only in its quotation of the definition of consensus, names no step for a sustained opposition as such, gives no criteria for an important part of the concerned interests, and has no step for registering an opposition (SD-4, whole document; searched: sustained, concerned interest, important part, register, record, minute, assess, address, resolve, opposition, objection, Against).

e. **Fact.** Neither SD-4 nor any ISO/IEC Directives Part 1 clause listed here limits the number of participants whose costs one sponsor pays (SD-4, whole document; ISO/IEC Directives Part 1, 0.7, 1.7, 1.8.2, 1.12.1 to 1.12.3, 2.1.6, 2.1.7, 2.5.6 to 2.5.10, 2.6.2 to 2.6.5, 2.7.2 to 2.7.7, 3.1, 5.3.1, 5.4 to 5.7, 6.1 to 6.5; searched: sponsor, employer, company, organization).

f. **Fact.** SD-4 assigns no one the role of ruling which side of a technical dispute is correct or of judging a design's quality (SD-4, whole document; searched: rule, ruling, decide, arbiter, correct, authority, quality, judge, evaluat, assess, merit).

g. **Fact.** SD-4 contains no requirement that a concern raised privately with a chair be recorded (SD-4, whole document; searched: privately, record, minute, concern).

h. **Fact.** Beyond showing the design alternatives considered and bringing new information when declining a directed change, SD-4 asks no proposal paper to state the objections raised against its proposal; it asks a TS or white paper for its exit criteria, "especially, the list of must-address controversial issues" (SD-4, whole document; searched: objection, concern, answer, address, rebut, respon, rationale, revision, feedback, oppos, disagree).

i. **Fact.** SD-4 asks no proposal paper to state its design's drawbacks, and of the searched terms it uses only "cost", for travel costs (SD-4, whole document; searched: trade-off, tradeoff, trade off, cost, drawback, downside, disadvantage, limitation, weakness).

j. **Fact.** SD-4 asks no author to state who pays their costs or what interests they or a sponsor have in a proposal (SD-4, whole document; searched: disclos, declar, affiliat, interest, employ, sponsor, fund, compan, behalf).

k. **Fact.** SD-4 names no step in which anyone checks whether designs adopted in separate papers work together (SD-4, whole document; searched: interact, integrat, coheren, consisten, together, harmon, conflict, fits).

l. **Fact.** SD-4 asks no paper that proposes part of a design to state what later parts will require of it, and names no step for reviewing such parts together (SD-4, whole document; searched: part, parts, later, future, rule out, preclud, foreclos, extension, extend, subsequent, follow-on, builds on, depend, together).

In plain terms: the proof starts from assumptions about cost, time, preference, standing, learning, and memory, from defined terms that include two kinds of author, and from the rules as SD-4 and the ISO/IEC Directives state them, in the edition of the Directives that replaces the JTC 1 Supplement. For the United States position it also uses INCITS's rules for counting votes.

2. Who decides, and with what moves

59. **Proposition.** The participants present decide a design's content: chairs call consensus on their votes, counted under the 2:1 norm, in the design rooms and the study groups feeding them. SD-4 assigns no one to judge a design's quality, wording rooms check only that wording expresses the design, and national bodies ballot only on whole documents. *[from 43b, 43f, 43r, 44d, 44e, 43k, 43m, 58f, 43s, 43u, 43t, 54h, 47b, 54b, 24]*
60. **Proposition.** Each person in a subgroup's room casts one vote, guests a convenor invites included, while no user votes in any WG21 poll or in the ballots on its drafts. *[from 43b, 52j, 52a, 54h, 18, 23, 25]*
61. **Proposition.** The players are result-seeking authors, design-bound authors, regulars, study group chairs, design room chairs, the convenor, national bodies, sponsors, and users. The proof takes as given the choices of wording room chairs, guests, participants who are not regulars, and the direction group, and the convenor's appointments, study groups, and guest invitations. *[from 59, 60, 34, 21, 37, 38, 19, 24, 25, 22, 23]*
62. **Proposition.** Each player has these moves under the rules. *[proved by 62a, 62b, 62c, 62d, 62e, 62f, 62g]*
    a. **Lemma.** An author can submit papers, including papers that each propose part of a design, and present them or have a qualified and prepared presenter do so. An author can also revise them and ask the chair for a poll or the design room for a ship vehicle, which the chair and the design rooms decide. *[from 42a, 42h, 42j, 43c, 43m, 21]*
    b. **Lemma.** A regular can attend or not, study a paper or not, speak, and vote Favor, Neutral, or Against, or not at all. A regular can also write a paper, raise a concern privately with a chair, and ask their national body to take a national position. *[from 1, 6, 43n, 43b, 43g, 42a, 42g, 43o, 49b, 19]*
    c. **Lemma.** A chair can choose the polls, order the agenda by SD-4's priorities, time-box a new proposal and poll whether to go on, re-poll after discussion, ask Against voters for reasons, and allow limited exploration of alternatives. A design room or wording room chair also makes the consensus call that forwards a proposal, a design room chair can delay a proposal one meeting for a rival without a paper, and a study group chair polls which design leaves the group as its initial proposal. *[from 43c, 43v, 43i, 42d, 43p, 43f, 43h, 42l, 43r, 43u, 42k, 43k, 24]*
    d. **Lemma.** The convenor can appoint chairs, form study groups, invite a guest for up to two meetings, and make plenary consensus calls. In plenary the convenor can ask whether Against votes are personal or national body positions. *[from 43a, 43l, 52j, 44d, 44b, 44c]*
    e. **Lemma.** A national body can appoint experts, vote Yes or No or abstain on ballots, comment, state a national position in plenary, and appeal. It can also object in SC22 and JTC 1 to action taken on a topic not on the agenda. *[from 52e, 54h, 52a, 51f, 47d, 44c, 42e, 56a]*
    f. **Lemma.** A sponsor can pay for participation and choose whom and how many it pays, and a category C liaison's representatives act for it. Unless it is an ISO member body or a liaison organization, a sponsor has no vote of its own in WG21's polls or in the ballots on WG21's drafts. *[from 22, 2, 58e, 43b, 52e, 52f, 54h, 52a, 25]*

g. **Lemma.** A user casts no vote in WG21's polls or ballots but can give feedback on a TS or white paper or write a paper a qualified and prepared participant presents. A user can also raise their interests with a national body, which ISO asks to take account of all interests concerned at the national level, or become a participant. *[from 60, 45b, 42h, 51e, 23]*

3. What a vote costs

63. **Lemma.** Under the 2:1 norm each Against vote raises by two the Favor votes a proposal needs, while Neutral votes and abstentions leave the count unchanged. One Against vote changes the result only at one or two votes above the line, and at least a third of the Favor-plus-Against votes cast Against keeps the norm unmet. *[proved by 63a, 63b, 63c, 63d]*
    a. **Lemma.** Under the 2:1 norm, each Against vote raises by two the number of Favor votes a proposal needs. *[from 43f, 26]*
    b. **Lemma.** Neutral votes and abstentions do not enter the 2:1 comparison, though SD-4 says a Neutral majority of those voting can be concerning. *[from 43f, 43g, 26]*
    c. **Lemma.** One more Against vote changes whether the 2:1 norm is met only when the Favor votes exceed twice the other Against votes by one or two. *[from 43f, 26]*
    d. **Lemma.** Any group casting at least one third of the Favor-plus-Against votes as Against prevents the 2:1 norm from being met. *[from 43f, 26]*
64. **Proposition.** For a regular, an Against vote on a colleague's proposal costs more than a Neutral vote or none. It risks the colleague's later support no less, and only Against voters can be asked for reasons, which take time to prepare. *[from 5, 3, 4, 7, 43h, 58b, 19, 27]*
65. **Proposition.** A regular's doubt about a colleague's proposal enters the 2:1 count only if it is worth the extra cost of an Against vote. That vote changes the result only when the count sits one or two votes above the line, though it can also win changes to the design. *[proved by 65a, 65b] [from 63, 64, 4, 43h, 43q, 28, 29]*
    a. **Lemma.** When an Against vote is worth less to the regular than its extra cost, the regular ranks a Neutral vote or none above it. *[from 64, 4, 29]*
    b. **Lemma.** When an Against vote is worth more to the regular than its extra cost, the regular ranks it above a Neutral vote or none. *[from 64, 4, 29]*

Table 1. One regular's vote on a colleague's proposal that the regular doubts. Columns: what an Against vote is worth to the regular, compared with its added cost. Each cell: the regular's rank of the vote. 2 is best within each column.

	Worth below cost	Worth above cost
Neutral or abstain	**2**	1
Against	1	**2**

[from 65a, 65b]

66. **Proposition.** With many Favor votes cast, one objector's Against vote changes the result only if enough others also vote Against, so its worth to each objector depends on the others' votes. The same objectors can then form either of two settled patterns: just enough of them vote Against to block, or none does. *[from 63, 65, 64, 28, 34]*
67. **Proposition.** A participant can argue for another design and still vote Neutral, which leaves the count unchanged and draws no request for reasons. *[from 43n, 63, 43h, 58b]*
68. **Proposition.** Two regulars with pending proposals each risk the other's support for their own by opposing the other's proposal. When each regular's doubt is worth less than that risk and the time for reasons, neither opposes, and both do better than if both opposed. *[proved by 68a, 68b] [from 5, 4, 64, 65, 63, 29]*
    a. **Lemma.** In the case the table below describes, each regular ranks refraining above opposing, whatever the other does. That holds because an Against vote on the other's proposal is then worth less to them than its extra cost. *[from 65, 65a, 5]*
    b. **Lemma.** In that case, each regular ranks opposing while the other refrains above refraining while the other opposes, since the other's opposition costs them more than their own opposing does. *[from 4, 29]*

Table 2. Two regulars, each with a proposal pending, neither of which lowers what the other values by more than the cost of opposing it, and for each of whom the other's opposition costs more than their own opposing does. Refrains means votes Favor, Neutral, or not at all. First number: A's rank. Second number: B's rank. 4 is best. † marks a Pareto-optimal cell.

	B refrains	**B opposes**
A refrains	**4, 4 †**	2, 3
A opposes	3, 2	1, 1

[from 68a, 68b]

4. What the record keeps and how objections count

69. **Proposition.** A proposal can advance in WG21's rooms and plenary with objections unresolved, since it needs a consensus call, which the 2:1 norm normally supports, not every objector's persuasion. *[proved by 69a, 69b]*
    a. **Lemma.** Under ISO, the obligation to address a sustained opposition is no obligation to resolve it, and a sustained opposition is not a veto. Still, ISO's general principles say consensus requires the resolution of substantial objections. *[from 53g, 51d]*

b. **Proposition.** A proposal can advance while discussed objections to it remain unresolved, because advancing requires a consensus call, which the 2:1 norm normally supports, and consensus need not imply unanimity. *[from 69a, 43r, 43u, 44d, 43f, 41a]*

70. **Proposition.** The record SD-4 and the searched ISO clauses require of a WG21 poll need not show whether an objection was answered or outvoted, or why consensus was called over it. *[proved by 70a, 70b, 70c] [from 58c]*
    a. **Lemma.** SD-4 requires minutes and lets poll questions and numeric results be quoted publicly, but requires no record of how an objection was answered or why a consensus call was made. Plenary may ask a subgroup chair to summarize the reason for a decision. *[from 50b, 50a, 58a, 44f]*
    b. **Lemma.** ISO lets participants comment on or object to minutes before they are confirmed, and has the chair of SC22 see that points of view at SC22's meetings are summarized. It bars participants from commenting publicly on others' contributions. *[from 55c, 52b, 55a]*
    c. **Lemma.** The chair of SC22 judges consensus at SC22's committee stage, and ISO 2.5.7 recommends steps for a sustained opposition, including registering it, to the leadership. It does not say whether a convenor applies them inside a working group, and under either reading SD-4 adds no step for assessing or registering an opposition in subgroup polls. *[from 53b, 53a, 53j, 53f, 53h, 53c, 58d, 25]*
71. **Proposition.** SD-4 adopts ISO's definition of consensus, which turns on sustained opposition, but names no step for judging whether an opposition is sustained or for registering one, and no criteria for an important part of the concerned interests. Whoever ISO's leadership is for a working group, SD-4's own steps are the 2:1 norm after discussion, reasons from Against voters on request, escalation, and plenary's 3-way poll and question about national positions. *[from 52c, 41a, 53c, 43f, 43h, 49a, 44b, 44c, 53f, 58d]*
72. **Lemma.** Under SD-4's definition, an objection prevents consensus only if it concerns a substantial issue, is maintained, and comes from an important part of the concerned interests. The leadership judges that part case by case, none of the three conditions asks whether the objection is right, and every view must still be heard. *[from 41a, 53c, 53d, 41b, 40f]*
73. **Proposition.** When an objection's Against votes are too few to stop the 2:1 norm, the cheapest response keeping it met at that poll is no change or a partial change. Finding out whether those objectors are right leaves that count unchanged, and SD-4 expects any change the room directs to be made or defended with new information. *[from 4, 11, 7, 63, 43q]*
74. **Proposition.** SD-4 asks an Against voter to be ready to name the changes that would win them over, or to say that none would. Concerns stated as changes give authors items to grant one at a time, while opposition in any form offers none and weighs in the poll only as a vote. *[from 43h, 62a]*
75. **Proposition.** Of minutes, non-public email lists, meeting records, ballot comments, and papers, only a paper puts an objection's argument outside the password-protected materials without anyone else's help. Writing one takes time. *[from 70, 50a, 55c, 47d, 49d, 49e, 62b, 7]*

76. **Proposition.** A proposal's case lasts for decades in its required paper, while an objection's argument lasts outside the protected records only in a paper its holder writes. A reason no written record holds survives only with those who heard it, and once they all stop attending, a later participant can voice it only by rediscovering it. *[from 42a, 13, 75, 70]*
77. **Proposition.** SD-4 describes an objector who presses on after a consensus call, without new information or allowed escalation, as not accepting a group decision, whether or not anyone answered it. It says the same of a national body comment repeating a decided objection, unless the comment brings new information or the body warned all along that it would vote No. *[from 41c, 40e, 41d, 41e, 47g, 47h, 47i]*
78. **Proposition.** SD-4 gives an objection weight at plenary only if its holder escalated it by 5pm the evening before, on the email lists and within their national body. There, by default, it is not a national position, and after the deadline even new information goes to the email lists or a ballot comment, with a discussion paper. *[from 49c, 49a, 49b, 49d, 49e]*
79. **Proposition.** No design is considered on merit alone: a proposal needs an on-time paper and a presenter, a rival needs a paper to exist, and objections need fixed forms and times. Participants learn these rules partly by attending, some only that way, and other things equal, whoever knows them has the better chance of adoption, whatever their design skill. *[from 42f, 42h, 42k, 43b, 78, 47h, 12, 62]*
80. **Proposition.** Other things equal, a newcomer, lacking what earlier meetings teach, spends more effort getting a paper considered than a returning participant, though SD-4 expects everyone to know its procedures. Where WG21 runs study groups that incubate early proposals, such as EWGI and LEWGI, a newcomer's early proposal can start there. *[from 12, 20, 39b, 43k, 24]*

5. Momentum: incumbents and the cost of reversal

81. **Induction.** Each decision built on a design adds work that reversing the design would redo or discard, and more is built after each favorable poll. So the cost of reversing a design grows as it advances, whether or not the design is wrong. *[from 10, 4, 33, 43r, 43u, 44d]*
82. **Proposition.** A design passing a direction poll becomes the incumbent, and a rival without a paper cannot block it, though the chair may delay it one meeting for one to arrive. A rival with a paper can be polled alongside it, the room can direct changes, and by default a TS merge proceeds with it unless an alternative is actively progressing. *[from 42a, 42k, 42l, 43q, 43d, 42n, 31, 30]*
83. **Proposition.** A design can become the incumbent at its first direction poll, when it has had less review than at any later meeting. From then on, every rival and every later change to it bears a reversal cost that grows as more is built on it. *[from 9, 8, 31, 30, 82, 81, 33]*
84. **Proposition.** A rival wins a voter from an incumbent only with an advantage worth more to that voter than their share of the reversal's cost. That hurdle grows with each decision built on the incumbent, which faces no such hurdle itself. *[from 4, 81, 82, 31, 33, 29]*

85. **Lemma.** A direction poll asks voters only to agree that work should continue, less than a forwarding poll asks, so it can pass where a forwarding poll would fail. Yet passing it makes the design the incumbent. *[from 30, 31, 43r, 4, 83]*
86. **Proposition.** If a poll places a proposal among the work the committee aims to include in the next IS, the proposal joins a category SD-4 has chairs take up first. A meeting's hours are fixed, so the time it then gets comes out of the time left for papers outside those categories. *[from 43v, 14]*
87. **Proposition.** A longer, more complex design costs more to study and has more places where a defect can go unnoticed. Yet a reviewer whose stake does not grow with its size does better studying it less closely than a smaller design with the same stake. *[from 7, 6, 8, 4, 28, 29]*
88. **Proposition.** Participants who do not study a proposal typically abstain, so one costing more to study than most stakes in it is voted on mainly by those with large stakes. Unless those of them who object cast at least a third of the Favor-plus-Against votes, it can advance with no change at all. *[from 43g, 87, 7, 4, 63, 73]*
89. **Proposition.** A design normally needs the 2:1 norm to advance, and once adopted, a paper or ballot comment to remove it normally needs the same, switching the default to keeping it. After a successful final ballot, design changes are expected to have near-unanimous consent, ISO bars technical changes at approval, and SD-4 calls published features much harder to update incompatibly. *[from 43f, 42a, 47e, 47j, 54c, 45g, 33]*
90. **Proposition.** An objector who arrives after a design has been adopted faces three obstacles at once. Removal needs a paper meeting the 2:1 norm, reversal costs more for every decision built on the design, and the record keeps its case but unwritten objections only in memory. *[from 89, 81, 84, 76, 78]*

6. Sponsors, proposal size, and parts

91. **Proposition.** A sponsor adds a voter to a room for each participant it pays to attend, and neither SD-4 nor the searched ISO clauses limit how many it pays. SC22 may cap only the experts each national body or liaison appoints, and at SC22 itself each national body counts once. *[from 60, 58e, 43b, 2, 52e, 52i, 52h, 53a]*
92. **Proposition.** Changing the features a sponsor's interests depend on costs that sponsor rework of what it built on them, lost fit with its requirements, and more review, all still to come. The gain of a better design is spread across all the feature's users, and the same 2:1 norm decides every proposal, however many users will rely on it. *[from 4, 10, 7, 17, 43f, 22]*
93. **Induction.** Each part of a design adopted in parts becomes work every later part builds on, so a rival to the whole design must replace every part already adopted. *[from 62a, 81, 33]*
94. **Proposition.** Presenting a design in parts lets the first parts be adopted before the whole is reviewed, and each adopted part adds to what a rival to the whole must replace. The cost is a paper and a presentation for each part. *[from 62a, 43r, 43u, 44d, 93, 84, 7, 42h]*
95. **Proposition.** Reviewers cannot study what later parts need of a first part that fixes a design's structure, SD-4 asks no paper to say, and once adopted, every rival must replace it. So a design-bound author's

best first paper is the smallest part that fixes that structure and still meets the 2:1 norm against opponents who foresee the rest. *[from 9, 6, 4, 28, 58l, 94, 93, 84, 63, 38]*

96. **Proposition.** A large proposal has more sponsored authors in the room than a small one, and more meetings where a poll can make it the incumbent or end that standing. Each step it advances brings it nearer to work that can make plenary progress, which SD-4 has chairs take up first. *[from 36, 35, 31, 91, 43v, 43r, 43u, 44d]*
97. **Proposition.** A small proposal's author bears every meeting's cost alone, and outside SD-4's priority categories the proposal gets agenda time only after the priority work. Where WG21 runs study groups that incubate early proposals, such as EWGI and LEWGI, SD-4's statement that study groups own incubating applies to them. *[from 35, 1, 43v, 42d, 43k, 24]*
98. **Proposition.** Each result about large or small proposals rests on an attribute their definitions name: who pays the authors' costs, the number of authors, review's expected span, or the paper's length. So it holds for any proposal with that attribute, whatever its size. *[from 35, 36, 96, 97]*

7. Study groups and deadlocks

99. **Proposition.** On a topic with a study group, rival designs are first polled there, among those who chose that room over the rooms meeting with it. The design it sends on is where the design room starts, and the design room, which owns selecting designs, can still poll a rival that has a paper. *[proved by 99a, 99b, 99c] [from 43m, 43d]*
    a. **Lemma.** SD-4 has the convenor form a study group on a design room chair's recommendation when a topic has broad interest, no unified initial proposal, and a strong chair candidate. It gives the study group the role of producing an initial proposal that can then enter the design room. *[from 43l, 43k]*
    b. **Lemma.** A participant can attend only one of the rooms meeting at the same time, and votes and speaks only there. So a study group's voters are those for whom its topic outweighs the rooms meeting with it. *[from 16, 4, 28, 43b]*
    c. **Lemma.** The design room starts from the study group's initial proposal, which a rival without a paper does not block while it is an active proposal making progress. *[from 99a, 99b, 42k, 43k]*
100. **Proposition.** Paying participants to attend a study group puts a sponsor's voters where the design the design room starts from is chosen. In a room with fewer voters, fewer paid participants make up a third of the Favor-plus-Against votes, enough to block, or over two thirds, enough to meet the norm. *[from 99, 91, 63]*
101. **Proposition.** When each of two groups casts at least a third of the Favor-plus-Against votes against the other's design, neither design meets the 2:1 norm. The deadlock lasts until voters change position, a design changes, the voters present change, a chair calls consensus below the norm, or the design room sends both to competing TSs. *[from 63, 43f, 43r, 43m, 54i]*

102. **Proposition.** In a long deadlock, a group whose sponsors pay its costs bears less per meeting than a group paying its own way. In the case the table below describes, the first holds and the second yields, whichever design is better. *[proved by 102a, 102b, 102c, 102d, 102e, 102f, 102g] [from 2, 1, 101]*
    a. **Lemma.** When one deadlocked group yields, the other holds, and no one else votes Against the holder's design, the deadlock ends with that design advancing. The holder gains its whole stake, the yielder none of its own, and neither bears the cost of holding through a deadlock. *[from 101, 63, 28]*
    b. **Lemma.** When both groups hold and the deadlock ends with either design equally likely, each can expect half its stake on average, less its cost of holding through it. Members whose costs sponsors pay bear less of that cost per meeting. *[from 101, 1, 2, 28]*
    c. **Lemma.** Each group does better holding while the other yields, gaining its whole stake at no cost of holding, than when both yield to a compromise worth half its stake. *[from 102a, 4, 29]*
    d. **Lemma.** Each group does better when both yield to a compromise worth half its stake than when both hold. That holds because a deadlock both hold brings half its stake on average, less a cost of holding. *[from 102b, 1, 4, 29]*
    e. **Lemma.** Each group does better when both yield to a compromise worth half its stake than when it yields while the other holds, which brings none of its stake. *[from 102a, 4, 29]*
    f. **Lemma.** A group whose cost of holding is less than half its stake does better holding through a deadlock both hold than yielding while the other holds. That holds because half its stake on average, less that cost, leaves it better off than none of its stake. *[from 102b, 102a, 4, 29]*
    g. **Lemma.** A group whose cost of holding is more than half its stake does better yielding while the other holds than holding through a deadlock both hold. That holds because half its stake on average, less that cost, leaves it worse off than none of its stake. *[from 102b, 102a, 4, 29]*

Table 3. Two groups in a deadlock, with equal stakes in their own designs. Group A's members have their costs paid by sponsors and group B's pay their own, so holding through the deadlock costs group A less than half its stake and group B more than half. A deadlock both hold ends with either design equally likely, and both yielding means a compromise worth half its stake to each. First number: group A's rank. Second number: group B's rank. 4 is best. † marks a Pareto-optimal cell.

	B holds	**B yields**
A holds	2, 1	**4, 2** †
A yields	1, 4 †	3, 3 †

[from 102c, 102d, 102e, 102f, 102g]

8. Chairs, the convenor, and plenary

103. **Proposition.** Every consensus call has a cost: declaring consensus over Against votes risks later national body opposition, and declining delays work SD-4 says subgroup polls should favor. Departing from the 2:1 norm can cost the caller's post: chairs serve at the convenor's choice with no fixed term, and the convenor for terms SC22 may decline to renew. *[from 43e, 44c, 54h, 49b, 62e, 43a, 52d, 4, 25]*
104. **Proposition.** Above the 2:1 line after discussion, declaring consensus follows SD-4's norm, though a Neutral majority can be concerning, and at or below it SD-4 allows a re-poll after discussion. A chair whose stake is worth less than what departing can cost makes the call the norm gives. *[proved by 104a, 104b] [from 43f, 63, 43h, 103]*
    a. **Lemma.** A design room chair whose stake is worth less than what departing can cost ranks declaring consensus above declining or re-polling on a count above 2:1 after discussion. This holds when Neutral votes are not a majority of those voting. *[from 43f, 103, 4, 29]*
    b. **Lemma.** On a count at or below 2:1 after discussion, such a chair ranks declining or re-polling above declaring consensus. *[from 43f, 103, 4, 29]*

Table 4. A design room chair's consensus call, when the chair's stake in the proposal is worth less to them than what departing from SD-4's norm can cost them, and Neutral votes are not a majority of those voting. Columns: the count after discussion of the Against voters' concerns. Each cell: the chair's rank of the call. 2 is best within each column.

	Above 2:1	Not above 2:1
Declare consensus	**2**	1
Decline or re-poll	1	**2**

[from 104a, 104b]

105. **Proposition.** Of two papers otherwise alike, a chair does better taking up first the one whose author has a record of following procedures and responding to review, which promises fewer problems. That record comes only from taking part. *[from 3, 4, 14, 43v, 29]*
106. **Proposition.** A participant who thinks a new proposal should not take up extended subgroup time can tell the chair privately and in advance. The chair may then time-box it and poll whether to go on, and nothing in SD-4 requires a record of the request. *[from 43o, 43p, 58g]*
107. **Proposition.** At plenary, design stays with the design rooms, objections not escalated in time get no weight, and all present are asked for serious design concerns. Most polls pass by unanimous consent, and when opposition is significant the convenor usually asks whether Against votes are personal or national body positions. *[from 44b, 44e, 78, 44a, 44c, 59]*

9. Users, quality, and disclosure

108. **Proposition.** No user votes, and taking part costs the participant or their sponsor, while a better design's gain is spread across all users and seldom turns on one user's effort. Users' interests reach decisions only through what participants themselves value, papers a participant presents, TS feedback, and national bodies, which form positions only where that is worth their members' time. *[from 60, 62g, 17, 4, 51e, 49b, 14]*
109. **Proposition.** The rules bar participation fees and exclude no registered expert, yet only those who bear the cost of taking part, or whose sponsors do, get a vote. They propose for needs they and their sponsors know, and since users differ from participants in which features matter, the needs reaching a vote need not match users' needs. *[from 52i, 55b, 51c, 51b, 47a, 60, 1, 2, 4, 17, 42h]*
110. **Proposition.** An author gains at adoption, while a flaw that implementation and use have not yet exposed costs users only after adoption, and can fall on any user of the feature. *[from 4, 8, 17, 21, 23]*
111. **Lemma.** Implementation and use reduce uncertainty about how well a design works, though even then a room cannot fully judge it at a poll. SD-4 describes a TS or white paper as a way to gain that experience, noting that its users typically should not rely on it in production code. *[from 8, 45a, 45b]*
112. **Proposition.** Winning a consensus call and producing the design that serves users best coincide only as far as voters can judge that at the poll and weigh it above other values. SD-4 assigns no one to judge it for them. *[from 4, 8, 59, 72, 111, 58f]*
113. **Proposition.** Close study costs the reviewer, while the gain of a better design is spread across all its users. So review worth more to users together than it costs, but less to any one participant, is review no participant does better doing. *[from 7, 17, 4, 87, 28, 29]*
114. **Proposition.** No step in SD-4 checks whether separately adopted designs work together, and each author gains from their own design's adoption while a conflict costs users and later rework. The nearest steps ask a good paper to fit the language's principles, have new wording written against the adopted drafts, and let a TS merge early for integration testing. *[from 58k, 42b, 46a, 45d, 4, 8, 17, 10]*
115. **Lemma.** SD-4 relies on the presenter, an expert in the paper's domain, to educate a room of general C++ experts. It also counts a feature's designers among the best experts on any concern about it. *[from 42i, 48a]*
116. **Proposition.** Stating a drawback in the paper before an early poll can cost Favor votes when incumbency is decided, while a better-informed vote's gain is spread across the design's users. SD-4 says a good paper should show the alternatives considered, but it asks no paper to state its design's drawbacks, and a paper is considered without them. *[from 6, 83, 115, 17, 4, 42c, 58i, 42f, 42h]*
117. **Proposition.** No rule requires a written answer to an objection, apart from an author's defense of declining a directed change and ISO's response to each enquiry ballot comment. For the author, a written answer makes the objection easier to raise again, but also shows later what would not be new, the test SD-4 applies to repeated objections. *[from 70, 58h, 43q, 54g, 54f, 13, 77, 47h]*

10. The schedule and national bodies

118. **Proposition.** A proposal that misses an edition's feature cutoff waits at least for the next edition, planned three years later. It can ship sooner as a white paper, which has no prescribed schedule, and the same schedule keeps an unfinished feature from delaying the features that are ready. *[proved by 118a] [from 45f, 44d, 15, 32]*
    a. **Lemma.** Once an edition's draft goes to ballot, SD-4 treats a ballot comment that requests adding a feature as out of scope, and WG21 plans the next edition three years later. *[from 47f, 32, 15]*
119. **Proposition.** At the feature cutoff, a one-meeting delay for a paperless rival, or an Against vote keeping a proposal below the 2:1 line, sends it to the next planned edition. *[from 118, 42k, 63, 44d, 32]*
120. **Proposition.** A TS delays IS adoption by about one to two years, and its experience can bring incompatible changes first: a gain to result-seeking authors, a risk to design-bound ones. The design room can still require a TS, and ISO and SC22 now recommend the lighter white paper. *[from 42m, 45a, 45b, 45g, 43m, 45e, 45f, 111, 37, 38, 4]*
121. **Lemma.** Every national body may vote on enquiry and final ballots, which pass when two thirds of the P-members' votes are positive and at most a quarter of all votes negative, not counting abstentions or negative votes without technical comments. ISO encourages a body without national consensus or expert input to abstain, sustained opposition does not apply to these votes, and a final draft that fails goes back to SC22. *[from 52a, 54a, 54h, 54b, 51f, 53e, 54d, 25]*
122. **Proposition.** A national body's No opposes every feature in the document at once, including ones its own experts support, while WG21 considers its comments whatever its vote. A body valuing the rest of a document above blocking one feature does better voting Yes with comments, and one wanting the work in no form does better voting No. *[proved by 122a, 122b, 122c, 122d] [from 59, 52f, 47c, 54g, 54e, 121, 4, 29]*
    a. **Lemma.** A body that values the rest of a document above blocking any one feature ranks Yes with comments above abstaining. That holds because only the Yes counts toward the approval it prefers, and its comments are considered either way. *[from 121, 47c, 4]*
    b. **Lemma.** Such a body ranks abstaining above No, since only the No counts against the approval it prefers and its comments gain nothing from a No. *[from 121, 47c, 4]*
    c. **Lemma.** A body that wants the work in no form ranks No above abstaining, since only the No counts against approval, which is what SD-4 says No is for. *[from 121, 47b, 4]*
    d. **Lemma.** Such a body ranks abstaining above Yes with comments, since only the Yes counts toward approval of work it wants in no form. *[from 121, 47b, 4]*

Table 5. A national body's ballot vote. Columns: how the body values the document; "Values the rest" means it values the rest of the document above blocking any one feature in it, and "Opposes the work" means it does not want the work at all. Each cell: its rank of the vote. 3 is best within each column.

	Values the rest	Opposes the work
Yes with comments	**3**	1
Abstain	2	2
No	1	**3**

[from 122a, 122b, 122c, 122d]

123. **Proposition.** At ballot, only a national body that values blocking one feature above the whole rest of the document does better voting No over it. SD-4 expects such a body to have told the committee all along, and ISO asks every national body to make its standpoint clear early. *[from 122, 47b, 47i, 41e, 51e, 4, 29]*
124. **Proposition.** A national body forms its position by its panel's own rules, which SD-4 does not state, so whether a sponsor's paid participants add weight there depends on those rules. In the United States, INCITS gives a voting member's representatives one vote, apart from up to four Defense Department votes, and bars consultants a voting member funds from voting membership. *[proved by 124a] [from 49d, 49b, 2]*
    a. **Lemma.** INCITS, whose Executive Board approves the United States position that goes to ANSI, gives each organization one voting membership, apart from up to four Department of Defense votes on technical committees, admits a consultant as a voting member only if no voting member funds it, and lets the representatives of standards developing organizations and consortia be employed by other voting members. A United States delegation may also decide a new issue at an international plenary by caucus consensus. *[from 57a, 57b, 57c, 57d, 57e]*

11. The settled pattern

125. **Lemma.** Whatever the others choose, a regular does no better than to vote Favor on colleagues' proposals they support and, on the rest, to vote Neutral or not at all unless a doubt is worth more than an Against vote's extra cost. That cost is the risk to colleagues' support and the time for reasons, and the vote changes the result only when the count sits one or two votes above the line. *[from 64, 65, 63, 66, 67, 68, 77, 119, 62, 34]*
126. **Lemma.** Whatever the others choose, a design-bound author does no better than to seek direction polls as early as they can pass, propose first the smallest part that fixes the features its gain depends on, and seek the IS over a TS unless the room requires one. It does no better than to leave out trade-offs no reviewer would raise, ask for no poll against a rival it expects to lose, and answer objections with the cheapest change that keeps the 2:1 norm met, leaving those features alone unless the room directs otherwise. *[from 83, 81, 84, 85, 95, 94, 120, 116, 73, 69, 74, 92, 43q, 31, 38, 62a, 34]*

127. **Lemma.** Whatever the others choose, a result-seeking author does no better than to state a trade-off and grant a change whenever the better design they can bring is worth more to it than the votes or work they cost. It does no better than to seek an early poll or an early first part only where sooner adoption outweighs the higher cost this puts on later improvements, and to back a rival that would serve users better by more than its share of the reversal's cost. *[from 83, 81, 84, 94, 116, 73, 74, 37, 62a, 34]*
128. **Lemma.** Between two options, an author who values both a good result for users and particular features chooses as the author type would whose value differs more between them. *[from 4, 37, 38, 29]*
129. **Proposition.** The rules treat both kinds of author alike: the same moves, and no request to state a design's drawbacks, a sponsor, or what the author's gain depends on. So stating a trade-off can cost a result-seeking author Favor votes that a design-bound author who leaves it out keeps. *[proved by 129a, 129b] [from 62a, 58i, 58j, 42f, 42h, 116, 126, 127]*
    a. **Lemma.** In the case the table below describes, a result-seeking author ranks stating the trade-off above leaving it out. That holds because stating it brings a design that serves users better while the proposal still meets the 2:1 norm. *[from 37, 116, 6, 4, 29]*
    b. **Lemma.** In that case, a design-bound author ranks leaving the trade-off out above stating it, since stating it brings a change to a feature its gain depends on. *[from 38, 116, 6, 4, 29]*

Table 6. An author's choice, before an early direction poll, whether to state in the paper a trade-off no reviewer would otherwise raise, when stating it would lead the room to change a feature a design-bound author's gain depends on, in a way that serves users better, and the proposal would meet the 2:1 norm either way. Columns: the author's type. Each cell: the author's rank of the choice. 2 is best within each column.

	Result-seeking	Design-bound
State it	**2**	1
Leave it out	1	**2**

[from 129a, 129b]

130. **Lemma.** Whatever the others choose, a small proposal's author does no better than to keep it presented, in person, remotely where offered, or by another qualified and prepared presenter, while it is worth more to them than costs they bear alone. When those costs, which grow with each meeting it waits behind priority work, outweigh it, the author does no better than to let it stop. *[from 97, 80, 1, 51c, 42h, 42f, 62, 34]*
131. **Lemma.** Whatever the others choose, a study group chair does no better than to poll the competing designs, side by side where there are several, and send on a design that meets the 2:1 norm, or none when none does. The chair does better departing from this only when their stake in the contest is worth more to them than what departing can cost them. *[from 99, 101, 102, 103, 43l, 43k, 43d, 43f, 62, 34]*

132. **Lemma.** Whatever the others choose, a design room chair does no better than to declare consensus above the 2:1 line after discussion, when Neutral votes are not a majority, and to re-poll or decline at or below it, unless their stake is worth more than what departing can cost them. It does no better than to take up first, of papers otherwise alike, the one whose author has a record of following the procedures. *[from 104, 103, 105, 106, 85, 119, 123, 62, 34]*
133. **Lemma.** Whatever the others choose, the convenor does no better than to follow SD-4's plenary procedure unless their stake is worth more than what departing can cost them. That procedure has the convenor declare consensus when no one objects to unanimous consent, or when all opposition rests on objections not escalated in time and holds no national position. *[from 107, 78, 103, 44b, 49c, 44d, 41a, 62, 34]*
134. **Lemma.** Whatever the others choose, a national body does no better than to vote Yes with comments if it values a document's rest above blocking any one feature, and No otherwise. It does no better than to raise its objections early, as SD-4 and ISO ask, and ISO encourages it to abstain when it lacks national consensus or expert input. *[from 122, 123, 121, 124, 77, 41e, 51e, 62, 34]*
135. **Lemma.** Whatever the others choose, a sponsor does no better than to pay for participants in the rooms whose decisions it values, a study group included. That holds as long as the votes they add are worth their cost to it. *[from 91, 100, 96, 124, 1, 2, 62, 34]*
136. **Lemma.** Whatever the others choose, a user does no better than to take the move, if any, with the largest positive net gain: taking part, TS feedback, a paper a participant presents, or a request to a national body. Taking part costs one person the full price while a better design's gain is shared by all users, so it is that move only for a user whose own stake outweighs that cost. *[from 108, 62g, 1, 17, 4, 29, 34]*
137. **Proposition.** The rules check these effects: every view must be heard, a third of the votes can block, rivals with papers can be polled together, and implementation experience informs votes. Plenary invites serious design concerns, national bodies vote on each balloted draft and have enquiry comments answered, editing committees verify applied changes, minutes can be challenged, and appeals are open. *[proved by 137a, 137b, 137c, 137d, 137e]*
    a. **Lemma.** SD-4 requires every view to be heard and understood and the majority to listen, and asks for respect, good faith, and no collusion, with conduct concerns reported through ISO's process. It welcomes negative comments, and any group casting a third of the Favor-plus-Against votes as Against keeps a proposal from meeting the 2:1 norm. *[from 40f, 41b, 43n, 40d, 40b, 40c, 40a, 63]*
    b. **Lemma.** Chairs may allow limited exploration of alternatives, and SD-4 encourages polling alternatives side by side and says a good paper shows the alternatives considered and its design principles. Followup and rebuttal papers can be considered, and an author who declines a directed change should defend that decision. *[from 42l, 43d, 42c, 42b, 42g, 43q]*
    c. **Lemma.** Competing TSs are possible, a TS or white paper should list the controversial issues to address before it merges, and SD-4 expects pre-merge issues to be addressed. Implementation experience informs voters, a white paper has no prescribed schedule, and the planned schedule keeps an unfinished feature from delaying ready ones. *[from 54i, 45h, 45c, 111, 45f, 118]*

d. **Lemma.** Every expert is appointed by a national body or liaison organization, SC22 may cap how many each appoints, and ISO recommends that experts keep their national body informed. ISO asks national bodies to take account of all interests concerned nationally, SD-4 asks design rooms to weigh direction group opinions on any proposal, and agenda topics need on-time papers. *[from 52e, 52h, 52g, 51e, 43j, 42e]*

e. **Lemma.** SD-4 asks those with a concern to tell the chairs and authors early, plenary solicits serious design concerns, every national body may vote on each enquiry and final ballot, every enquiry comment gets a written response, and an editing committee verifies that approved changes were applied correctly. Minutes can be challenged, appeals are open, on drafts only on questions of principle or harm to the reputation of ISO or IEC, ISO says consensus requires resolving substantial objections and ends the committee stage when all technical issues are resolved, and SD-4 works in addition to the Directives. *[from 48b, 44a, 121, 54f, 54g, 46b, 55c, 56a, 56b, 56c, 56d, 53i, 51d, 53j, 39a]*

138. **Theorem.** Under SD-4 and the quoted clauses of ISO/IEC Directives Part 1, in the edition replacing the JTC 1 Supplement, and given the axioms, these choices form a settled pattern. In it, no player improves their position by changing only their own choice. A design-bound author seeks direction polls as early as they can pass, proposes first the smallest part fixing the features it needs, and seeks the IS over a TS. It leaves out trade-offs no reviewer would raise and answers objections with the cheapest change that keeps the 2:1 norm met. A result-seeking author states trade-offs and makes changes whenever the better design they bring is worth their cost to it. It backs a rival that serves users better by more than its share of the reversal's cost. An author who values both chooses, at each two-way choice, as the type whose value differs more. A small proposal's author keeps it presented while it is worth costs borne alone. A regular votes Favor on colleagues' proposals they support and otherwise Neutral or not at all, voting Against only for a doubt worth more than that vote's extra cost. A sponsor pays for participants where the decisions it values are made, study groups included. Chairs and the convenor follow SD-4's procedure unless their stake outweighs what departing can cost them. A national body votes Yes with comments unless blocking one feature is worth as much to it as the rest of the document. A user takes part only when their own stake outweighs a cost borne alone, and otherwise uses a cheaper move or none.

    In this pattern a design's content can be fixed when it is least reviewed. Its first favorable poll can make it the incumbent, and every later change or rival then pays a reversal cost that grows whether or not the design is wrong. Objections count only in the forms and times that attending teaches. In a long deadlock, the group whose sponsors pay its costs can hold longest, whichever design is better. A proposal that costs more to study than most participants' stakes in it can advance with no change at all. An objector who arrives after adoption meets these obstacles together. Other things equal, a proposal advances more readily when its authors keep presenting it or know the procedures, or when a sponsor pays for participants who hold its position. The same holds when the chair expects fewer problems from it, a poll has placed it among the work aimed at the next IS, or it is already the incumbent. No user votes, and the needs that reach a vote are those participants and their sponsors

know, which need not match those of users at large. The participants present decide content, and a sponsor can add voters without limit. A change to a design costs the sponsors who have built on it, while its gain is spread across all users. An author gains at adoption, while a hidden flaw costs users later. Technical merit enters through what participants value, including a good result for users, and through the discussion, papers, and implementation experience that inform their votes. SD-4 assigns no one to rule on it, and the consensus test never asks whether an objection is right. No step checks whether separately adopted designs work together, review whose gain is spread across all users goes undone, and the record need not show whether objections were answered. The rules ask no author to state a trade-off. Leaving one out costs a design-bound author nothing the rules impose, while stating one can cost a result-seeking author votes. A first part that fixes a structure is reviewed before anyone can study what the later parts need of it, and every rival must then replace it. The rules also contain checks on each of these effects, and each result about large or small proposals holds for any proposal with the attributes it rests on. *[from 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 59, 82, 83, 81, 84, 79, 88, 90, 86, 105, 102, 108, 109, 91, 92, 72, 110, 112, 111, 114, 113, 70, 116, 95, 98, 61, 51a, 58f, 42h, 12, 4, 34]*

12. Corollaries

139. **Corollary.** If the settled pattern holds, the record should show the following. SD-4 keeps meeting records password-protected but lets poll questions and numeric results be quoted, and each prediction names its evidence, counter-evidence, an innocent explanation, and the data telling them apart. *[proved by 139a, 139b, 139c, 139d, 139e, 139f, 139g, 139h, 139i, 139j, 139k] [from 50a]*
    a. **Corollary.** Where one sponsor pays both a proposal's authors and many of the participants present, and its interests include the proposal's adoption, Against votes are a smaller share of the Favor-plus-Against votes than on other proposals; poll results and stated affiliations, taken as the observable measure of who pays, are the evidence, and an Against share no lower would count against it. An innocent explanation is that colleagues share technical views, and comparing their votes with their votes on proposals of similar design on the same topic from authors with other affiliations, whose adoption the sponsor's interests do not include, would tell the two apart. *[from 91, 4, 65, 63, 138]*
    b. **Corollary.** Among proposals with similar stakes, the share of those present not voting rises with paper length; poll results, attendance, and lengths are evidence, and no rise would count against it. An innocent explanation is that longer papers cover narrower topics fewer participants know, and comparing papers on topics of similar breadth but different length would tell the two apart. *[from 87, 43g, 138]*

c. **Corollary.** Large proposals appear as sequences of papers, each building on parts already adopted; the mailings are the evidence, and large designs adopted in one paper would count against it. An innocent explanation is that large designs are easier to review in parts, and whether parts are published together or each only after the last is adopted tells them apart. *[from 93, 94, 95, 96, 138]*

d. **Corollary.** The first papers of designs later extended part by part describe their scope as small and seldom set out the later parts they serve; the mailings are the evidence, and first papers that set out their later parts would count against it. An innocent explanation is that reviewers prefer small steps and a small first step does lower risk, and checking whether later parts relied on choices in the first that it did not describe would tell the two apart. *[from 95, 116, 126, 138]*

e. **Corollary.** After a design passes a direction poll, papers and poll questions on its topic present rivals as changes to it, even when no technical content has changed; poll questions and paper titles are the evidence, and rivals presented as equal alternatives would count against it. An innocent explanation is that the wording tracks the incumbent's real change of status, and comparing whether rivals polled as changes drew fewer Favor votes than comparable rivals polled side by side would tell the two apart. *[from 82, 83, 84, 138]*

f. **Corollary.** Removals after adoption are rare, can come through a paper or a national body comment, and are more likely when national bodies whose No votes could fail a ballot oppose the design; working drafts, ballot comments, and poll results are the evidence, and frequent removals by narrow margins would count against it. An innocent explanation is that adopted designs rarely need removal, and comparing removals with the defect reports filed against adopted features would tell the two apart. *[from 89, 81, 122, 121, 138]*

g. **Corollary.** Small proposals stop advancing without a negative poll more often than large proposals do; the mailings and poll results are the evidence, and no such difference would count against it. An innocent explanation is that small proposals are more often dropped on reflection, and comparing self-funded and sponsored authors' stops on papers of similar length would tell the two apart. *[from 130, 97, 96, 138]*

h. **Corollary.** National bodies state positions on few proposals; national body comments and plenary records of national positions are the evidence, and positions on most proposals would count against it. An innocent explanation is that national bodies agree with most proposals, and comparing how often they state positions on proposals with and without Against votes in subgroups tells them apart. *[from 78, 108, 49b, 14, 121, 138]*

i. **Corollary.** Papers that advance over Against votes often leave unstated the objections they faced, beyond the alternatives and declined directions SD-4 asks papers to address; the papers and poll results are the evidence, and papers that routinely state and answer their objections would count against it. An innocent explanation is that the objections were answered in discussion, and comparing such papers with the answers subgroup chairs give at plenary when asked would tell the two apart. *[from 70, 117, 58h, 7, 138]*

j. **Corollary.** If some objections left unresolved point to real defects, features adopted while Against votes remained show more defects found after adoption than features adopted by unanimous consent; issues lists, correction papers, removals, and poll results are the evidence, and no difference would count against it. An innocent explanation is that contested features are more novel or more used, so more people look for their defects, and comparing features matched on scope, novelty, implementation experience before adoption, and use would tell the two apart. *[from 69, 73, 112, 8, 138]*

k. **Corollary.** If closer study before adoption finds more defects, topics on which competing papers from separate author groups reached the deciding poll show fewer defects after adoption than topics with one paper; author lists, mailings, issues lists, and removals are the evidence, and no difference would count against it. An innocent explanation is that topics drawing competing papers are the important or well-understood ones, which draw expert attention anyway, and comparing topics matched on scope and novelty would tell the two apart. *[from 87, 113, 4, 43d, 138]*

140. **Corollary.** Each change below, to a rule or to what a move costs, would remove the support for a named item, and the model predicts the stated effect. *[proved by 140a, 140b, 140c, 140d, 140e, 140f, 140g, 140h, 140i]*

a. **Corollary.** If design polls were anonymous, the standing part of an Against vote's cost would shrink for voters who do not speak against the proposal. The chair could no longer ask Against voters for reasons, and the model predicts more Against votes on colleagues' proposals. *[from 64, 5, 65, 138]*

b. **Corollary.** If a chair could ask voters of every position for reasons, Neutral would bear the reasons cost Against now bears, while not voting would bear none. The model predicts fewer Neutral votes and more participants not voting. *[from 64, 63, 43g, 138]*

c. **Corollary.** If the rules limited how many participants whose costs one sponsor pays may vote in a room, the votes a sponsor can add would be capped. The model predicts smaller differences in poll margins between large and small proposals. *[from 91, 96, 135, 138]*

d. **Corollary.** If the rules required a written answer to each objection in the record, the record would show whether an objection was answered. The model predicts that papers advancing over Against votes would state the objections they faced. *[from 70, 117, 58h, 138]*

e. **Corollary.** If the rules set a decision procedure for deadlocks lasting a set number of meetings, a deadlock lasting that long would end by that procedure. The model predicts fewer deadlocks that end when one side stops attending. *[from 102, 101, 138]*

f. **Corollary.** If a TS did not delay adoption compared with an edition, the author's cost of accepting a TS would fall. The model predicts more designs sent to a TS or white paper. *[from 120, 138]*

g. **Corollary.** If SD-4 adopted ISO's steps for assessing, registering, and addressing a sustained opposition, with a recorded answer, it would state a step for assessing and registering one. The model predicts registered oppositions in WG21's records. *[from 71, 70, 138]*

h. **Corollary.** If forming national positions during meetings cost less, national bodies would take more positions, and more significant plenary opposition would hold national positions, which the room's count does not settle. The model predicts larger changes in answer to objections national bodies back, slower progress for contested proposals, and less weight at plenary for how many participants each side has present. *[from 78, 107, 133, 134, 73, 91, 124, 138]*

i. **Corollary.** If forming national positions during meetings cost more, national bodies would take fewer, plenary would seldom meet opposition holding national positions, and the whole-document ballot would be their only check. The model predicts that attendance, paid participants, and early incumbency would decide still more of each design's content. *[from 59, 123, 133, 134, 91, 83, 138]*